



```
1- def merge_sort(arr):
2-     if len(arr) > 1:
3-         mid = len(arr) // 2
4-         left = arr[:mid]
5-         right = arr[mid:]
6-
7-         merge_sort(left)
8-         merge_sort(right)
9-
10-        i = j = k = 0
11-
12-        while i < len(left) and j < len(right):
13-            if left[i] < right[j]:
14-                arr[k] = left[i]
15-                i += 1
16-            else:
17-                arr[k] = right[j]
18-                j += 1
19-            k += 1
20-
21-        while i < len(left):
22-            arr[k] = left[i]
23-            i += 1
24-            k += 1
25-
26-        while j < len(right):
27-            arr[k] = right[j]
28-            j += 1
29-            k += 1
30-
31- arr = [38, 27, 43, 3, 9, 82, 10]
32- merge_sort(arr)
33- print("Sorted Array:", arr)
```

Sorted Array: [3, 9, 10, 27, 38, 43, 82]

=== Code Execution Successful ===